SALT ANALYSIS I - LEAD ACETATE - ($CH_3COO)_2Pb$

EXPERIMENT	OBSERVATION	INFERENCE
PRELIMINARY TESTS :- **1. COLOUR** Colour of the given salt is noted.	Colourless	Shows the absence of Cu^{2+},Fe^{2+}.Fe^{3+},Ni^{2+}, Mn^{2+}, Co^{2+}
2.SMELL Smell of the given salt is noted.	Vinegar like smell	Shows the presence of acetate ion (CH_3COO^-)
3. Dry heating test:- Small amount of the salt is taken in a dry test and heated. On heating some salts undergo decomposition and evolve gases with characteristic colour and smell.	Acetic acid vapours with vinegar-like smell.	Shows the presence of acetate ion (CH_3COO^-)
4.Cobalt nitrate test:- A filter paper is soaked into the solution of salt,cobalt nitrate and conc.Nitric acid in a test tube and burnt in the non luminous part of the flame.	No characteristic colored residue.	Shows the absence of Zn^{2+}, Al^{3+} and Mg^{2+}
5.Flame test:- Small amount of the salt is taken in a watch glass and made into paste with conc.Hydrochloric acid then introduced into the non luminous part of the flame with the help of a glass stirrer.	Dull-bluish-white flame	Shows the presence of Pb^{2+}.

Identification of the acid radicals

EXPERIMENT	OBSERVATION	INFERENCE
6.Dilute sulphuric acid test:- Small amount of the salt is taken in a dry test and dilute sulphuric acid is added.	No characteristic gas is evolved.	Shows the absence of CO_3^{2-},SO_3^{2-},NO_2^- and S^{2-}
7. Concentrated sulphuric acid test:- Small quantity of salt with conc. sulphuric acid is taken in a test tube. The gas evolved in cold and then on heating is identified.	Colourless acetic acid vapours with vinegar like smell and turns blue litmus red.	Shows the presence of acetate ion (CH_3COO^-)

Confirmation of acid radicals

Preparation of sodium carbonate extract (SCE):-

A pinch of salt is mixed with double the amount of sodium carbonate and is boiled with distilled water for sometime. The suspension obtained is filtered.The filtrate is called sodium carbonate extract.

Confirmation of Acetate, CH_3COO^-

EXPERIMENT	OBSERVATION	INFERENCE
1.Ester Test :- Small amount of the salt in a test tube is heated with conc.sulphuric acid and ethanol. The reaction mixture is transferred into a beaker containing distilled water .	Pleasant fruity smell of ester.	Presence of acetate ion (CH_3COO^-) Is confirmed

2.Ferric chloride test:- To the salt solution in a test tube neutral ferric chloride solution is added and then filtered. The filtrate is divided into two portions. (i) To one part dil.HCl is added. (ii) To the second part water is added and boiled.	Reddish coloured filtrate is obtained. Reddish colour disappears. Reddish brown precipitate is formed.	Presence of acetate ion ($CH_3COO^{-)}$ is confirmed.

Identification of basic radicals

Preparation of original solution:-

Small amount of the salt is taken in a test tube and distilled water is added. Shaken well . The clear salt solution is called the original solution.

Analysis of group zero (NH_4^+):-

The solid salt is heated with a concentrated solution of sodium hydroxide.	No characteristic ammoniacal smell.	Absence of group zero (NH_4^+)

GROUP SEPARATION

Analysis of group I:- To the original salt solution in a test tube dil.Hydrochloric acid is added.	White precipitate is obtained.	Presence of group I (Pb^{2+})

CONFIRMATION OF BASIC RADICAL - LEAD (Pb^{2+})

1.Potassium Iodide test:- To the original salt solution in a test tube Potassium iodide solution is added.	Yellow precipitate is formed which dissolves in boiling water and on cooling recrystallises.	Presence of lead (Pb^{2+}) is confirmed.
2. Potassium chromate test:- To the original salt solution in a test tube Potassium chromate solution is added.	Yellow precipitate is formed which dissolves in sodium hydroxide solution.	Presence of lead (Pb^{2+}) is confirmed

Result

1. The acid radical present in the given salt is Acetate - CH_3COO^-
2. The basic radical present in the given salt is Lead - Pb^{2+}
3. The given salt is Lead acetate - $(CH_3COO)_2Pb$

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SALT ANALYSIS - 2 Aluminium Nitrate ($Al(NO_3)_3$)

EXPERIMENT	OBSERVATION	INFERENCE
PRELIMINARY TESTS :- **1. COLOUR**		

Colour of the given salt is noted. **2.SMELL** Smell of the given salt is noted. **3. Dry heating test:-** Small amount of the salt is taken in a dry test and heated. On heating some salts undergo decomposition and evolve gases with characteristic colour and smell. **4.Cobalt nitrate test:-** A filter paper is soaked into the solution of salt,cobalt nitrate and conc.Nitric acid in a test tube and burnt in the non luminous part of the flame. **5.Flame test:-** Small amount of the salt is taken in a watch glass and made into paste with conc.Hydrochloric acid then introduced into the non luminous part of the flame with the help of a glass stirrer.	Colourless No characteristic odour Reddish brown fumes of NO2 gas which turns ferrous sulphate solution black. Blue colour residue is obtained. No characteristic flame is observed.	Shows the absence of Cu^{2+},Fe^{2+}.Fe^{3+},Ni^{2+}, Mn^{2+}, Co^{2+} Shows the absence of NH_4^+, CH_3COO^- and S^{2-} Shows the presence of NO_3^- Shows the presence of Al^{3+} Shows the absence of Ca^{2+},Ba^{2+},Sr^{2+},Cu^{2+}, Zn^{2+}, Pb^{2+} etc

Identification of the acid radicals

6.Dilute sulphuric acid test:- Small amount of the salt	No characteristic gas has	Shows the absence of

is taken in a dry test and dilute sulphuric acid is added. **7. Concentrated sulphuric acid test:-** Small quantity of salt with conc. sulphuric acid is taken in a test tube. The gas evolved in cold and then on heating is identified.	evolved. No characteristic gas has evolved	CO_3^{2-},SO_3^{2-},NO_2^- and S^{2-} Shows the absence Cl^-,Br^-, I^-,NO_3^- and CH_3COO^-

Confirmation of acid radicals

Preparation of sodium carbonate extract (SCE):-

A pinch of salt is mixed with double the amount of sodium carbonate and is boiled with distilled water for sometime. The suspension obtained is filtered.The filtrate is called sodium carbonate extract.

Confirmation of Sulphate (SO_4^{2-})

1.Barium chloride test: To the aqueous solution of the salt in a test tube , barium chloride solution is added.	White precipitate is formed which is insoluble in dil.HCl	Presence of sulphate(SO_4^{2-}) is confirmed.
2. Lead acetate test: To the aqueous solution of the salt in a test tube ,lead acetate solution is added.	White precipitate is formed which is soluble in excess of hot ammonium acetate solution	Presence of sulphate(SO_4^{2-}) is confirmed.

Identification of basic radicals

Preparation of original solution:-

Small amount of the salt is taken in a test tube and distilled water is added. Shaken well . The clear salt solution is called the original solution.

Analysis of group zero (NH_4^+)

The solid salt is heated with concentrated solution of sodium hydroxide.	No characteristic ammoniacal smell.	Absence of group zero (NH_4^+)

GROUP SEPARATION

Analysis of group I To the original salt solution in a test tube dil.Hydrochloric acid is added.	No white precipitate is obtained.	Absence of group I (Pb^{2+})
Analysis of group II To the original solution in a test tube, H_2S gas is passed in the presence of dil. HCl	No characteristic precipitate is obtained.	Absence of group II (Cu^{2+} , As^{2+})

Analysis of group III To the original solution in the test tube	White gelatinous	Presence of group III

ammonium hydroxide (NH_4OH) is added in the presence of ammonium chloride (NH_4Cl).	precipitate is obtained.	(Al^{3+} , Fe^{3+})

CONFIRMATION OF BASIC RADICAL - Aluminium (Al^{3+})

1. Lake test The white gelatinous precipitate is dissolved in dil.HCl. Two drops of blue litmus solution are added. Ammonium hydroxide is added in drops till blue colour develops.	Blue precipitate floating in the colourless solution is obtained.	Presence of Aluminium (Al^{3+}) is confirmed.
2.Cobalt nitrate test A filter paper is soaked into the solution of salt,cobalt nitrate and conc.Nitric acid in a test tube and burnt in the non luminous part of the flame.	Blue colour residue is obtained.	Presence of Aluminium (Al^{3+}) is confirmed.

Result

1. The acid radical present in the given salt is Sulphate (SO_4^{2-})
2. The basic radical present in the given salt is Aluminium (Al^{3+}).
3. The given salt is Aluminium Sulphate ($Al_2(SO_4^{2-})_3$)

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SALT ANALYSIS - 3 Zinc sulphate (ZnSO4)

EXPERIMENT	OBSERVATION	INFERENCE
PRELIMINARY TESTS :- **1. COLOUR** Colour of the given salt is noted.	Colourless	Shows the absence of Cu^{2+},Fe^{2+}.Fe^{3+},Ni^{2+}, Mn^{2+}, Co^{2+}
2.SMELL Smell of the given salt is noted.	No characteristic odour	Shows the absence of NH_4^+, CH_3COO^- and S^{2-}
3. Dry heating test:- Small amount of the salt is taken in a dry test and heated. On heating some salts undergo decomposition and evolve gases with characteristic colour and smell.	Colour of the residue is yellow when hot and white when cold is obtained.	Shows the presence of Zn^{2+}
4.Cobalt nitrate test:- A filter paper is soaked into the solution of salt,cobalt nitrate and conc.Nitric acid in a test tube and burnt in the non luminous part of the flame.	Green colored residue.	Shows the presence of Zn^{2+}.
5.Flame test:- Small amount of the salt is taken in a watch glass and made into paste with conc.Hydrochloric acid then introduced into the non luminous part of the flame with the help of a glass stirrer.	Green flashes	Shows the presence of Zn^{2+}

Identification of acid radicals

6.Dilute sulphuric acid test:- Small amount of the salt is taken in a dry test and dilute sulphuric acid is added.	No characteristic gas has evolved.	Shows the absence of CO_3^{2-},SO_3^{2-},NO_2^- and S^{2-}
7. Concentrated sulphuric acid test:- Small quantity of salt with conc. sulphuric acid is taken in a test tube. The gas evolved in cold and then on heating is identified.	No characteristic gas is evolved.	Shows the absence Cl^-,Br^-, I^-,NO_3^- and CH_3COO^-

Confirmation of acid radicals

Preparation of sodium carbonate extract (SCE):-

A pinch of salt is mixed with double the amount of sodium carbonate and is boiled with distilled water for sometime. The suspension obtained is filtered.The filtrate is called sodium carbonate extract.

Confirmation of Sulphate (SO_4^{2-})

1.Barium chloride test: To the aqueous solution of the salt in a test tube , barium chloride solution is added.	White precipitate is formed which is insoluble in dil.HCl.	Presence of sulphate(SO_4^{2-}) is confirmed.
2. Lead acetate test: To the aqueous solution of the salt in a test tube ,lead	White precipitate is formed which is soluble in excess of	Presence of sulphate(SO_4^{2-}) is confirmed.

acetate solution is added.	hot ammonium acetate solution.	

Identification of basic radicals

Preparation of original solution:-

Small amount of the salt is taken in a test tube and distilled water is added. Shaken well . The clear salt solution is called the original solution.

Analysis of group zero (NH_4^+)

The solid salt is heated with concentrated solution of sodium hydroxide.	No characteristic ammoniacal smell.	Absence of group zero (NH_4^+)

GROUP SEPARATION

Analysis of group I To the original salt solution in a test tube dil.Hydrochloric acid is added.	No white precipitate is obtained.	Absence of group I (Pb^{2+})
Analysis of group II To the original solution in a test tube,H2S gas is passed in the presence of dil. HCl.	No characteristic precipitate is obtained.	Absence of group II (Cu^{2+} , As^{2+})

Analysis of group III The original solution in the test tube	No characteristic	Absence of group III

ammonium hydroxide (NH_4OH) is added in the presence of ammonium chloride (NH_4Cl). **Analysis of group IV** The original solution in the test tube H2S gas is passed in the presence of ammonium chloride (NH_4Cl) and ammonium hydroxide(NH_4OH).	precipitate is obtained. Dull white precipitate is obtained.	(Al^{3+} , Fe^{3+}) Presence of group IV (Co^{2+},Mn^{2+},Ni^{2+} and Zn^{2+})

CONFIRMATION OF BASIC RADICAL - Zinc (Zn^{2+})

1. Sodium hydroxide test: To the original salt solution in the test tube sodium hydroxide solution is added in drops.	White precipitate is formed which is soluble in excess of sodium hydroxide solution.	Presence of Zinc (Zn^{2+}) is confirmed.
2. Potassium ferrocyanide test: To the original salt solution in the test tube potassium ferrocyanide solution is added.	White precipitate is obtained.	Presence of Zinc (Zn^{2+}) is confirmed

Result

1. The acid radical present in the given salt is Sulphate (SO_4^{2-})
2. The basic radical present in the given salt is Zinc (Zn^{2+}).
3. The given salt is Zinc sulphate ($ZnSO_4$)

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SALT ANALYSIS → 4 Barium chloride ($BaCl_2$)

EXPERIMENTS	OBSERVATION	INFERENCE
PRELIMINARY TESTS :- **1. COLOUR** Colour of the given salt is noted.	Colourless	Shows the absence of Cu^{2+},Fe^{2+}.Fe^{3+},Ni^{2+}, Mn^{2+}, Co^{2+}

2.SMELL Smell of the given salt is noted.	No characteristic odour	Shows the absence of NH_4^+, CH_3COO^- and S^{2-}
3. Dry heating test:- Small amount of the salt is taken in a dry test and heated. On heating some salts undergo decomposition and evolve gases with characteristic colour and smell.	Hydrogen chloride gas is evolved with pungent smell and gives white dense fumes with the rod dipped in ammonia solution.	Shows the presence of Cl^-
4.Cobalt nitrate test:- A filter paper is soaked into the solution of salt,cobalt nitrate and conc.Nitric acid in a test tube and burnt in the non luminous part of the flame.	No characteristic colored residue.	Shows the absence of Zn^{2+}, Al^{3+} and Mg^{2+}
5.Flame test:- Small amount of the salt is taken in a watch glass and made into paste with conc.Hydrochloric acid then introduced into the non luminous part of the flame with the help of a glass stirrer.	Persistent grassy - green colour flame (appearing after prolonged heating)	Shows the presence of Barium (Ba^{2+})

Identification of acid radicals

6.Dilute sulphuric acid test:- Small amount of the salt is taken in a dry test and dilute sulphuric acid is	No characteristic gas is evolved.	Shows the absence of CO_3^{2-},SO_3^{2-},NO_2^- and S^{2-}

added. **7. Concentrated sulphuric acid test:-** Small quantity of salt with conc. sulphuric acid is taken in a test tube. The gas evolved in cold and then on heating is identified.	Hydrogen chloride gas is evolved with a pungent smell and gives white dense fumes with the rod dipped in ammonia solution.	Shows the presence of Cl^-

Confirmation of acid radicals

Preparation of sodium carbonate extract (SCE):-

A pinch of salt is mixed with double the amount of sodium carbonate and is boiled with distilled water for sometime. The suspension obtained is filtered.The filtrate is called sodium carbonate extract.

Confirmation of Chloride (Cl^-)

1. Silver nitrate test: To the aqueous solution or sodium carbonate extract dil HNO3 is added. Boiled cooled and then silver nitrate solution is added.	White precipitate is obtained which is soluble in ammonium hydroxide solution .	Presence of chloride (Cl^-) is confirmed.
2. Chromyl chloride test: To a small amount of the salt in the test tube a pinch potassium	Red orange vapours(CrO_2Cl_2) evolved.	Presence of chloride (Cl^-) is confirmed.

dichromate ($K_2Cr_2O_7$) and conc.H_2SO_4 are added and heated. On passing the vapours into the test tube containing sodium hydroxide solution.	It turns yellow due to the formation of (Na_2CrO_4).	
On adding dil.acetic acid and lead acetate to the yellow solution.	Yellow precipitate is obtained .	

Identification of basic radicals

Preparation of original solution:-

Small amount of the salt is taken in a test tube and distilled water is added. Shaken well . The clear salt solution is called the original solution.

Analysis of group zero (NH_4^+)

The solid salt is heated with concentrated solution of sodium hydroxide.	No characteristic ammoniacal smell.	Absence of group zero (NH_4^+)

GROUP SEPARATION

Analysis of group I To the original salt solution in a test tube dil.Hydrochloric acid	No white precipitate is obtained.	Absence of group I (Pb^{2+})

is added. **Analysis of group II** To the original solution in a test tube,H2S gas is passed in the presence of dil. HCl.	No characteristic precipitate is obtained.	Absence of group II (Cu^{2+} , As^{2+})

Analysis of group III The original solution in the test tube ammonium hydroxide (NH_4OH) is added in the presence of ammonium chloride (NH_4Cl). **Analysis of group IV** To the original solution in the test tube H2S gas is passed in the presence of ammonium chloride (NH_4Cl) and ammonium hydroxide(NH_4OH). **Analysis of group V** To the original salt solution in a test tube ammonium carbonate [$(NH_4)_2CO_3$] is added in the presence of	No characteristic precipitate is obtained. No characteristic precipitate is obtained. White precipitate is obtained.	Absence of group III (Al^{3+} , Fe^{3+}) Absence of group IV (Co^{2+},Mn^{2+},Ni^{2+} and Zn^{2+}) Presence of group V (Ca^{2+},Ba^{2+} and Sr^{2+})

ammonium chloride (NH_4Cl) and ammonium hydroxide(NH_4OH).		

CONFIRMATION OF BASIC RADICALS - Barium (Ba2+)

1. Potassium chromate test: To the original salt solution in the test tube potassium chromate solution is added in drops.	Yellow precipitate is obtained.	Presence of Barium (Ba^{2+}) is confirmed.
2. Flame test: Small amount of the salt is taken in a watch glass and made into paste with conc.Hydrochloric acid then introduced into the non luminous part of the flame with the help of a glass stirrer.	Persistent grassy - green colour flame (appearing after prolonged heating)	Presence of Barium (Ba^{2+}) is confirmed.

Result

1. The acid radical present in the given salt is Chloride (Cl^-)
2. The basic radical present in the given salt is Barium (Ba^{2+}).
3. The given salt is Barium chloride ($BaCl_2$)

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SALT ANALYSIS → 5 Calcium chloride ($CaCl_2$)

EXPERIMENTS	OBSERVATION	INFERENCE
PRELIMINARY TESTS :- **1. COLOUR** Colour of the given salt is noted.	Colourless	Shows the absence of Cu^{2+},Fe^{2+}.Fe^{3+},Ni^{2+}, Mn^{2+}, Co^{2+}
2.SMELL Smell of the given salt is noted.	No characteristic odour	Shows the absence of NH_4^+, CH_3COO^- and S^{2-}
3. Dry heating test:- Small amount of the salt is taken in a dry test and heated. On heating some salts undergo decomposition and evolve gases with characteristic colour and smell.	Hydrogen chloride gas is evolved with pungent smell and gives white dense fumes with the rod dipped in ammonia solution.	Shows the presence of Cl^-
4.Cobalt nitrate test:- A filter paper is soaked into the solution of salt,cobalt nitrate and conc.Nitric acid in a test tube and burnt in the non luminous part of the flame.	No characteristic colored residue.	Shows the absence of Zn^{2+}, Al^{3+} and Mg^{2+}
5.Flame test:- Small amount of the salt is taken in a watch glass and made into paste with conc.Hydrochloric acid then introduced into the non luminous part of the flame with the help of a glass stirrer.	Brick red colour flame (Not-persistent)	Shows the presence of Calcium (Ca^{2+})

Identification of acid radicals

6.Dilute sulphuric acid test:- Small amount of the salt is taken in a dry test and dilute sulphuric acid is added.	No characteristic gas is evolved.	Shows the absence of CO_3^{2-},SO_3^{2-},NO_2^- and S^{2-}
7. Concentrated sulphuric acid test:- Small quantity of salt with conc. sulphuric acid is taken in a test tube. The gas evolved in cold and then on heating is identified.	Hydrogen chloride gas is evolved with pungent smell and gives white dense fumes with the rod dipped in ammonia solution.	Shows the presence of Cl^-

Confirmation of acid radical

Preparation of sodium carbonate extract (SCE):-

A pinch of salt is mixed with double the amount of sodium carbonate and is boiled with distilled water for sometime. The suspension obtained is filtered.The filtrate is called sodium carbonate extract.

Confirmation of Chloride (Cl^-)

1. Silver nitrate test: To the aqueous solution or sodium carbonate extract dil HNO3 is added. Boil,cool and then a silver nitrate solution is added.	White precipitate is obtained which is soluble in ammonium hydroxide solution .	Presence of chloride (Cl^-) is confirmed.

2. Chromyl chloride test: To a small amount of the salt in the test tube a pinch potassium dichromate ($K_2Cr_2O_7$) and conc.H_2SO_4 are added and heated.	Red orange vapours($CrO_2Cl_{2)}$ evolved.	Presence of chloride ($Cl^{-)}$ is confirmed.
On passing the vapours into the test tube containing sodium hydroxide solution.	It turns yellow due to the formation of (Na_2CrO_4).	
On adding dil.acetic acid and lead acetate to the yellow solution.	Yellow precipitate is obtained .	

Identification of basic radicals

Preparation of original solution:-

Small amount of the salt is taken in a test tube and distilled water is added. Shaken well . The clear salt solution is called the original solution.

Analysis of group zero (NH_4^+)

The solid salt is heated with concentrated solution of sodium hydroxide.	No characteristic ammoniacal smell.	Absence of group zero (NH_4^+)

GROUP SEPARATION

Analysis of group I To the original salt solution in a test tube dil.Hydrochloric acid is added.	No white precipitate is obtained.	Absence of group I (Pb^{2+})
Analysis of group II To the original solution in a test tube,H2S gas is passed in the presence of dil. HCl.	No characteristic precipitate is obtained.	Absence of group II (Cu^{2+} , As^{2+})

Analysis of group III The original solution in the test tube ammonium hydroxide (NH_4OH) is added in the presence of ammonium chloride (NH_4Cl).	No characteristic precipitate is obtained.	Absence of group III (Al^{3+} , Fe^{3+})
Analysis of group IV To the original solution in the test tube H2S gas is passed in the presence of ammonium chloride (NH_4Cl) and ammonium hydroxide(NH_4OH).	No characteristic precipitate is obtained.	Absence of group IV (Co^{2+},Mn^{2+},Ni^{2+} and Zn^{2+})
Analysis of group V To the original salt	White precipitate is	Presence of group V

solution in a test tube ammonium carbonate [$(NH_4)_2CO_3$] is added in the presence of ammonium chloride (NH_4Cl) and ammonium hydroxide(NH_4OH).	obtained.	(Ca^{2+},Ba^{2+} and Sr^{2+})

CONFIRMATION OF BASIC RADICAL - Calcium (Ca2+)

1. Ammonium oxalate test: To the original salt solution in the test tube ammonium oxalate and ammonium hydroxide solutions are added.	white precipitate is obtained.	Presence of Calcium (Ca^{2+}) is confirmed.
2. Flame test: Small amount of the salt is taken in a watch glass and made into paste with conc.Hydrochloric acid then introduced into the non luminous part of the flame with the help of a glass stirrer.	Brick red colour flame (Not-persistent)	Presence of Calcium (Ca^{2+}) is confirmed.

Result

1. The acid radical present in the given salt is Chloride (Cl^-)

2. The basic radical present in the given salt is Calcium (Ca^{2+}).

3. The given salt is Calcium chloride ($CaCl_2$)

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SALT ANALYSIS - 6 Magnesium sulphate (MgSO4)

EXPERIMENT	OBSERVATION	INFERENCE
PRELIMINARY TESTS :- **1. COLOUR** Colour of the given salt is noted.	Colourless	Shows the absence of Cu^{2+},Fe^{2+}.Fe^{3+},Ni^{2+}, Mn^{2+}, Co^{2+}
2.SMELL Smell of the given salt is noted.	No characteristic odour	Shows the absence of NH_4^+, CH_3COO^- and S^{2-}
3. Dry heating test:- Small amount of the salt is taken in a dry test and heated. On heating some salts undergo decomposition and evolve gases with characteristic colour and smell.	White residue , glows on heating.	Shows the presence of Ba^{2+},Ca2+, Sr2+, Mg2+ and Al3+
4.Cobalt nitrate test:- A filter paper is soaked into the solution of salt,cobalt nitrate and conc.Nitric acid in a test tube and burnt in the non luminous part of the flame.	Pink colour residue	Shows the presence of Mg^{2+}
5.Flame test:- Small amount of the salt is taken in a watch glass and made into paste with conc.Hydrochloric acid	No characteristic coloured flame	Shows the absence of Zn^{2+}, Ca2+, Ba2+ , Cu2+, Sr2+ , Pb2+

then introduced into the non luminous part of the flame with the help of a glass stirrer.		

Identification of acid radicals

6.Dilute sulphuric acid test:- Small amount of the salt is taken in a dry test and dilute sulphuric acid is added.	No characteristic gas has evolved.	Shows the absence of CO_3^{2-},SO_3^{2-},NO_2^- and S^{2-}
7. Concentrated sulphuric acid test:- Small quantity of salt with conc. sulphuric acid is taken in a test tube. The gas evolved in cold and then on heating is identified.	No characteristic gas has evolved.	Shows the absence Cl^-,Br^-, I^-,NO_3^- and CH_3COO^-

Confirmation of acid radicals

Preparation of sodium carbonate extract (SCE):-

A pinch of salt is mixed with double the amount of sodium carbonate and is boiled with distilled water for sometime. The suspension obtained is filtered.The filtrate is called sodium carbonate extract.

Confirmation of Sulphate (SO_4^{2-})

1.Barium chloride test: To the aqueous solution of the salt in a test tube , barium chloride solution is	White precipitate is formed which is insoluble in dil.HCl.	Presence of sulphate(SO_4^{2-}) is confirmed.

added. 2. Lead acetate test: To the aqueous solution of the salt in a test tube ,lead acetate solution is added.	White precipitate is formed which is soluble in excess of hot ammonium acetate solution.	Presence of sulphate(SO_4^{2-}) is confirmed.

Identification of basic radicals

Preparation of original solution:-

Small amount of the salt is taken in a test tube and distilled water is added. Shaken well . The clear salt solution is called the original solution.

Analysis of group zero (NH_4^+)

The solid salt is heated with concentrated solution of sodium hydroxide.	No characteristic ammoniacal smell.	Absence of group zero (NH_4^+)

GROUP SEPARATION

Analysis of group I To the original salt solution in a test tube dil.Hydrochloric acid is added.	No white precipitate is obtained.	Absence of group I (Pb^{2+})
Analysis of group II To the original solution in a test tube,H2S gas is passed in the presence of dil. HCl.	No characteristic precipitate is obtained.	Absence of group II (Cu^{2+} , As^{2+})

Analysis of group III The original solution in the test tube ammonium hydroxide (NH_4OH) is added in the presence of ammonium chloride (NH_4Cl).	No characteristic precipitate is obtained.	Absence of group III (Al^{3+} , Fe^{3+})
Analysis of group IV To the original solution in the test tube H2S gas is passed in the presence of ammonium chloride (NH_4Cl) and ammonium hydroxide(NH_4OH).	No characteristic precipitate is obtained.	Absence of group IV (Co^{2+},Mn^{2+},Ni^{2+} and Zn^{2+})
Analysis of group V To the original salt solution in a test tube ammonium carbonate [$(NH_4)_2CO_3$] is added in the presence of ammonium chloride (NH_4Cl) and ammonium hydroxide(NH_4OH).	No characteristic precipitate is obtained.	Absence of group V (Ca^{2+},Ba^{2+} and Sr^{2+})
Analysis of group VI To the original salt	White precipitate is	Presence of group VI

solution in the test tube disodium hydrogen phosphate solution is added.	obtained.	(Mg^{2+})

CONFIRMATION OF BASIC RADICAL - Magnesium (Mg^{2+})

1. Ammonium Phosphate test: To the original salt solution in the test tube solid ammonium chloride and ammonium hydroxide are added in slight excess and then ammonium phosphate solution is added.	White precipitate is obtained.	Presence of Magnesium (Mg^{2+}) is confirmed.
2. Magneson test: To the original salt solution in the test tube magneson reagent is added.	Bluish green precipitate is formed.	Presence of Magnesium (Mg^{2+}) is confirmed.

Result

1. The acid radical present in the given salt is Sulphate (SO_4^{2-})
2. The basic radical present in the given salt is Magnesium (Mg^{2+}).
3. The given salt is Magnesium sulphate ($MgSO_4$)

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SALT ANALYSIS → 7 Ammonium phosphate $(NH_4)_3PO_4$

EXPERIMENTS	OBSERVATION	INFERENCE
PRELIMINARY TESTS :- **1. COLOUR** Colour of the given salt is noted.	Colourless	Shows the absence of Cu^{2+},Fe^{2+}.Fe^{3+},Ni^{2+}, Mn^{2+}, Co^{2+}
2.SMELL Smell of the given salt is noted.	Ammoniacal smell	Shows the presence of NH_4^+.
3. Dry heating test:- Small amount of the salt is taken in a dry test and heated. On heating some salts undergo decomposition and evolve gases with characteristic colour and smell.	Ammonia gas is evolved with characteristic odour and gives white dense fumes with rod dipped in dil.HCl brought near the mouth of the test tube	Shows the presence of NH_4^+
4.Cobalt nitrate test:- n A filter paper is soaked into the solution of salt,cobalt nitrate and conc.Nitric acid in a test tube and burnt in the non luminous part of the flame.	No characteristic colored residue.	Shows the absence of Zn^{2+}, Al^{3+} and Mg^{2+}
5.Flame test:- Small amount of the salt is taken in a watch glass and made into paste with conc.Hydrochloric acid then introduced into the non luminous part of the flame with the help of a glass stirrer.	No characteristic coloured flame.	Shows the absence of Ca^{2+}, Ba^{2+} , Cu^{2+}, Sr^{2+} , Pb^{2+}, and Zn^{2+}

Identification of acid radicals

6.Dilute sulphuric acid test:- Small amount of the salt is taken in a dry test and dilute sulphuric acid is added.	No characteristic gas is evolved.	Shows the absence of CO_3^{2-},SO_3^{2-},NO_2^- and S^{2-}
7. Concentrated sulphuric acid test:- Small quantity of salt with conc. sulphuric acid is taken in a test tube. The gas evolved in cold and then on heating is identified.	No characteristic gas is evolved.	Shows the absence Cl^-,Br^-, I^-,NO_3^- and CH_3COO^-

Confirmation of acid radicals

Preparation of sodium carbonate extract (SCE):-

A pinch of salt is mixed with double the amount of sodium carbonate and is boiled with distilled water for sometime. The suspension obtained is filtered.The filtrate is called sodium carbonate extract.

Confirmation of Phosphate (PO_4^{3-})

1.Ammonium molybdate test: To the aqueous solution of the salt or sodium carbonate extract in a test tube conc.Nitric acid is added and boiled. To this ammonium molybdate is added	Yellow precipitate is obtained.	Presence of phosphate(PO_4^{3-}) is confirmed.

in excess and again boiled. 2. Magnesia mixture test: To the aqueous solution of the salt or sodium carbonate extract in a test tube magnesia mixture is added.	White precipitate is obtained.	Presence of phosphate(PO_4^{3-}) is confirmed.

Identification of basic radicals

Preparation of original solution:-

Small amount of the salt is taken in a test tube and distilled water is added. Shaken well . The clear salt solution is called the original solution.

Analysis of group zero (NH_4^+)

The solid salt is heated with a concentrated solution of sodium hydroxide.	1.Characteristic ammoniacal smell. 2. The gas gives white fumes when a glass rod dipped in dil. HCl is brought near the mouth of the test tube. 3. When the gas is passed through Nessler's reagent, it gives brown precipitate.	Presence of group zero (NH_4^+)

GROUP SEPARATION

Analysis of group I To the original salt solution in a test tube dil.Hydrochloric acid is added.	No white precipitate is obtained.	Absence of group I (Pb^{2+})
Analysis of group II To the original solution in a test tube,H2S gas is passed in the presence of dil. HCl.	No characteristic precipitate is obtained.	Absence of group II (Cu^{2+} , As^{2+})

Analysis of group III The original solution in the test tube ammonium hydroxide (NH_4OH) is added in the presence of ammonium chloride (NH_4Cl).	No characteristic precipitate is obtained.	Absence of group III (Al^{3+} , Fe^{3+})
Analysis of group IV To the original solution in the test tube H2S gas is passed in the presence of ammonium chloride (NH_4Cl) and	No characteristic precipitate is obtained.	Absence of group IV (Co^{2+},Mn^{2+},Ni^{2+} and Zn^{2+})

ammonium hydroxide(NH_4OH). **Analysis of group V** To the original salt solution in a test tube ammonium carbonate [$(NH_4)_2CO_3$] is added in the presence of ammonium chloride (NH_4Cl) and ammonium hydroxide(NH_4OH).	No characteristic precipitate is obtained.	Absence of group V (Ca^{2+},Ba^{2+} and Sr^{2+})
Analysis of group VI To the original salt solution in the test tube disodium hydrogen phosphate solution is added.	No characteristic precipitate is obtained.	Absence of group VI (Mg^{2+})

CONFIRMATION OF BASIC RADICAL - Ammonium (NH_4^+)

Nessler's reagent test: To the original salt solution in the test tube Nessler's reagent is added.	Reddish brown precipitate is obtained.	Presence of ammonium (NH_4^+) is confirmed.

Result

1. The acid radical present in the given salt is phosphate(PO_4^{3-})
2. The basic radical present in the given salt is ammonium (NH_4^+).

3. The given salt is ammonium phosphate $(NH_4)_3\ PO_4$

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SALT ANALYSIS → 8 Ammonium Oxalate $(NH_4)_2C_2O_4$

EXPERIMENTS	OBSERVATION	INFERENCE
PRELIMINARY TESTS :- **1. COLOUR** Colour of the given salt is noted.	Colourless	Shows the absence of Cu^{2+},Fe^{2+}.Fe^{3+},Ni^{2+}, Mn^{2+}, Co^{2+}
2.SMELL Smell of the given salt is noted.	Ammoniacal smell	Shows the presence of NH_4^+.
3. Dry heating test:- Small amount of the salt is taken in a dry test and heated. On heating some salts undergo decomposition and evolve gases with characteristic colour and smell.	Ammonia gas is evolved with characteristic odour and gives white dense fumes with rod dipped in dil.HCl brought near the mouth of the test tube	Shows the presence of NH_4^+
4.Cobalt nitrate test:- A filter paper is soaked into the solution of salt,cobalt nitrate and conc.Nitric acid in a test tube and burnt in the non luminous part of the flame.	No characteristic colored residue.	Shows the absence of Zn^{2+}, Al^{3+} and Mg^{2+}
5.Flame test:- Small amount of the salt is taken in a watch glass and made into paste with conc.Hydrochloric acid then introduced into the	No characteristic coloured flame.	Shows the absence of Ca^{2+}, Ba^{2+} , Cu^{2+}, Sr^{2+} , Pb^{2+}, and Zn^{2+}

non luminous part of the flame with the help of a glass stirrer.		

Identification of acid radicals

6.Dilute sulphuric acid test:- Small amount of the salt is taken in a dry test and dilute sulphuric acid is added.	No characteristic gas has evolved.	Shows the absence of CO_3^{2-},SO_3^{2-},NO_2^- and S^{2-}
7. Concentrated sulphuric acid test:- Small quantity of salt with conc. sulphuric acid is taken in a test tube. The gas evolved in cold and then on heating is identified.	No characteristic gas has evolved.	Shows the absence Cl^-,Br^-, I^-,NO_3^- and CH_3COO^-

Confirmation of acid radicals

Preparation of sodium carbonate extract (SCE):-

A pinch of salt is mixed with double the amount of sodium carbonate and is boiled with distilled water for sometime. The suspension obtained is filtered.The filtrate is called sodium carbonate extract.

Confirmation of Oxalate ($C_2O_4^{2-}$)

1.Calcium chloride test:- To the aqueous solution of the salt in a test tube dilute acetic acid is added and boiled off CO_2.. Calcium chloride solution is added.	White precipitate is formed which is soluble in dil.Nitric acid.	Presence of oxalate ($C_2O_4^{2-}$) is confirmed.

2. Potassium permanganate test:- To the pinch of salt in a test tube dil.sulphuric acid is added . The mixture is heated to 60-70 deg. Then a few drops of $KMnO_4$ solution is added.	Pink colour of $KMnO_4$solution is decolourised with the evolution of CO_2.	Presence of oxalate ($C_2O_4^{2-}$) is confirmed.

Identification of basic radicals

Preparation of original solution:-

Small amount of the salt is taken in a test tube and distilled water is added. Shaken well . The clear salt solution is called the original solution.

Analysis of group zero (NH_4^+)

The solid salt is heated with a concentrated solution of sodium hydroxide.	1.Characteristic ammoniacal smell. 2. The gas gives white fumes when a glass rod dipped in dil. HCl is brought near the mouth of the test tube. 3. When the gas is passed through Nessler's reagent, it gives brown	Presence of group zero (NH_4^+)

	precipitate.	

GROUP SEPARATION

Analysis of group I To the original salt solution in a test tube dil.Hydrochloric acid is added.	No white precipitate is obtained.	Absence of group I (Pb^{2+})
Analysis of group II To the original solution in a test tube,H2S gas is passed in the presence of dil. HCl.	No characteristic precipitate is obtained.	Absence of group II (Cu^{2+} , As^{2+})

Analysis of group III The original solution in the test tube ammonium hydroxide (NH_4OH) is added in the presence of ammonium chloride (NH_4Cl).	No characteristic precipitate is obtained.	Absence of group III (Al^{3+} , Fe^{3+})
Analysis of group IV To the original solution in the test tube H2S gas is passed in the presence of ammonium chloride (NH_4Cl) and	No characteristic precipitate is obtained.	Absence of group IV (Co^{2+},Mn^{2+},Ni^{2+} and Zn^{2+})

ammonium hydroxide(NH_4OH). **Analysis of group V** To the original salt solution in a test tube ammonium carbonate [$(NH_4)_2CO_3$] is added in the presence of ammonium chloride (NH_4Cl) and ammonium hydroxide(NH_4OH).	No characteristic precipitate is obtained.	Absence of group V (Ca^{2+},Ba^{2+} and Sr^{2+})
Analysis of group VI To the original salt solution in the test tube disodium hydrogen phosphate solution is added.	No characteristic precipitate is obtained.	Absence of group VI (Mg^{2+})

CONFIRMATION OF BASIC RADICAL - Ammonium (NH_4^+)

Nessler's reagent test: To the original salt solution in the test tube Nessler's reagent is added.	Reddish brown precipitate is obtained.	Presence of ammonium (NH_4^+) is confirmed.

Result

1. The acid radical present in the given salt is oxalate ($C_2O_4^{2-}$)
2. The basic radical present in the given salt is ammonium (NH_4^+).

3. The given salt is ammonium oxalate $(NH_4)_2\ C_2O_4$

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SALT ANALYSIS → 9 Ammonium carbonate $(NH_4)_2CO_3$

EXPERIMENTS	OBSERVATION	INFERENCE
PRELIMINARY TESTS :- **1. COLOUR** Colour of the given salt is noted.	Colourless	Shows the absence of Cu^{2+},Fe^{2+}.Fe^{3+},Ni^{2+}, Mn^{2+}, Co^{2+}
2.SMELL Smell of the given salt is noted.	Ammoniacal smell	Shows the presence of NH_4^+.
3. Dry heating test:- Small amount of the salt is taken in a dry test and heated. On heating some salts undergo decomposition and evolve gases with characteristic colour and smell.	Ammonia gas is evolved with characteristic odour and gives white dense fumes with rod dipped in dil.HCl brought near the mouth of the test tube	Shows the presence of NH_4^+
4.Cobalt nitrate test:- A filter paper is soaked into the solution of salt,cobalt nitrate and conc.Nitric acid in a test tube and burnt in the non luminous part of the flame.	No characteristic colored residue.	Shows the absence of Zn^{2+}, Al^{3+} and Mg^{2+}
5.Flame test:- Small amount of the salt is taken in a watch glass and made into paste with conc.Hydrochloric acid then introduced into the	No characteristic coloured flame.	Shows the absence of Ca^{2+}, Ba^{2+} , Cu^{2+}, Sr^{2+} , Pb^{2+}, and Zn^{2+}

non luminous part of the flame with the help of a glass stirrer.		

EXPERIMENTS	OBSERVATION	INFERENCE
PRELIMINARY TESTS :- **1. COLOUR** Colour of the given salt is noted.	Colourless	Shows the absence of Cu^{2+},Fe^{2+}.Fe^{3+},Ni^{2+}, Mn^{2+}, Co^{2+}
2.SMELL Smell of the given salt is noted.	Ammoniacal smell	Shows the presence of NH_4^+.
3. Dry heating test:- Small amount of the salt is taken in a dry test and heated. On heating some salts undergo decomposition and evolve gases with characteristic colour and smell.	Ammonia gas is evolved with characteristic odour and gives white dense fumes with rod dipped in dil.HCl brought near the mouth of the test tube	Shows the presence of NH_4^+
4.Cobalt nitrate test:- A filter paper is soaked into the solution of salt,cobalt nitrate and conc.Nitric acid in a test tube and burnt in the non luminous part of the flame.	No characteristic colored residue.	Shows the absence of Zn^{2+}, Al^{3+} and Mg^{2+}
5.Flame test:- Small amount of the salt is taken in a watch glass and made into paste with conc.Hydrochloric acid	No characteristic coloured flame.	Shows the absence of Ca^{2+}, Ba^{2+} , Cu^{2+}, Sr^{2+} , Pb^{2+}, and Zn^{2+}

then introduced into the non luminous part of the flame with the help of a glass stirrer.		

Identification of acid radicals

6.Dilute sulphuric acid test:- Small amount of the salt is taken in a dry test and dilute sulphuric acid is added.	No characteristic gas has evolved.	Shows the absence of CO_3^{2-},SO_3^{2-},NO_2^- and S^{2-}
7. Concentrated sulphuric acid test:- Small quantity of salt with conc. sulphuric acid is taken in a test tube. The gas evolved in cold and then on heating is identified.	No characteristic gas has evolved.	Shows the absence Cl^-,Br^-, I^-,NO_3^- and CH_3COO^-

Confirmation of acid radicals

Preparation of sodium carbonate extract (SCE):-

A pinch of salt is mixed with double the amount of sodium carbonate and is boiled with distilled water for sometime. The suspension obtained is filtered.The filtrate is called sodium carbonate extract.

Confirmation of carbonate (CO_3^{2-})

1.Dil.HCl test:- To the aqueous solution of the salt in a test tube dilute hydrochloric acid is added. 2.Magnesium sulphate test:-	Brisk effervescence with the evolution of carbon dioxide which turns lime water milky is observed .	Presence of carbonate (CO_3^{2-}) is confirmed.

To the aqueous solution of the salt in a test tube Magnesium sulphate solution is added.	White precipitate is formed in cold	Presence of oxalate (CO_3^{2-}) is confirmed.

Identification of basic radicals

Preparation of original solution:-

Small amount of the salt is taken in a test tube and distilled water is added. Shaken well . The clear salt solution is called the original solution.

Analysis of group zero (NH_4^+)

The solid salt is heated with a concentrated solution of sodium hydroxide.	1.Characteristic ammoniacal smell. 2. The gas gives white fumes when a glass rod dipped in dil. HCl is brought near the mouth of the test tube. 3. When the gas is passed through Nessler's reagent, it gives brown precipitate.	Presence of group zero (NH_4^+)

GROUP SEPARATION

Analysis of group I To the original salt solution in a test tube dil.Hydrochloric acid is added.	No white precipitate is obtained.	Absence of group I (Pb^{2+})
Analysis of group II To the original solution in a test tube,H2S gas is passed in the presence of dil. HCl.	No characteristic precipitate is obtained.	Absence of group II (Cu^{2+} , As^{2+})

Analysis of group III The original solution in the test tube ammonium hydroxide (NH_4OH) is added in the presence of ammonium chloride (NH_4Cl).	No characteristic precipitate is obtained.	Absence of group III (Al^{3+} , Fe^{3+})
Analysis of group IV To the original solution in the test tube H2S gas is passed in the presence of ammonium chloride (NH_4Cl) and ammonium hydroxide(NH_4OH).	No characteristic precipitate is obtained.	Absence of group IV (Co^{2+},Mn^{2+},Ni^{2+} and Zn^{2+})

Analysis of group V To the original salt solution in a test tube ammonium carbonate [$(NH_4)_2CO_3$] is added in the presence of ammonium chloride (NH_4Cl) and ammonium hydroxide(NH_4OH).	No characteristic precipitate is obtained.	Absence of group V (Ca^{2+},Ba^{2+} and Sr^{2+})
Analysis of group VI To the original salt solution in the test tube disodium hydrogen phosphate solution is added.	No characteristic precipitate is obtained.	Absence of group VI (Mg^{2+})

CONFIRMATION OF BASIC RADICAL - Ammonium (NH_4^+)

Nessler's reagent test: To the original salt solution in the test tube Nessler's reagent is added.	Reddish brown precipitate is obtained.	Presence of ammonium (NH_4^+) is confirmed.

Result

1. The acid radical present in the given salt is carbonate (CO_3^{2-})
2. The basic radical present in the given salt is ammonium (NH_4^+).
3. The given salt is ammonium carbonate $(NH_4)_2\ CO_3$

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SALT ANALYSIS → 10 Lead Nitrate (Pb(NO3)2)

EXPERIMENTS	OBSERVATION	INFERENCE
PRELIMINARY TESTS :- **1. COLOUR** Colour of the given salt is noted.	Colourless	Shows the absence of Cu^{2+},Fe^{2+}.Fe^{3+},Ni^{2+}, Mn^{2+}, Co^{2+}
2.SMELL Smell of the given salt is noted.	No characteristic smell is observed.	Shows the absence of NH_4^+.CH_3COO^- and S^{2-}
3. Dry heating test:- Small amount of the salt is taken in a dry test and heated. On heating some salts undergo decomposition and evolve gases with characteristic colour and smell.	Reddish brown fumes of NO2 gas which turns ferrous sulphate solution black.	Shows the presence of NO_3^-
4.Cobalt nitrate test:- A filter paper is soaked into the solution of salt,cobalt nitrate and conc.Nitric acid in a test tube and burnt in the non luminous part of the flame.	No characteristic colored residue.	Shows the absence of Zn^{2+}, Al^{3+} and Mg^{2+}
5.Flame test:- Small amount of the salt is taken in a watch glass and made into paste with conc.Hydrochloric acid then introduced into the non luminous part of the	Dull-bluish-white flame	Shows the presence of Pb^{2+}.

flame with the help of a glass stirrer.		

Identification of acid radicals

6.Dilute sulphuric acid test:- Small amount of the salt is taken in a dry test and dilute sulphuric acid is added.	No characteristic gas has evolved.	Shows the absence of CO_3^{2-},SO_3^{2-},NO_2^- and S^{2-}
7. Concentrated sulphuric acid test:- Small quantity of salt with conc. sulphuric acid is taken in a test tube. The gas evolved in cold and then on heating is identified.	Reddish brown fumes of NO2 gas with pungent smell , turns ferrous sulphate solution black	Shows the presence of NO_3^-

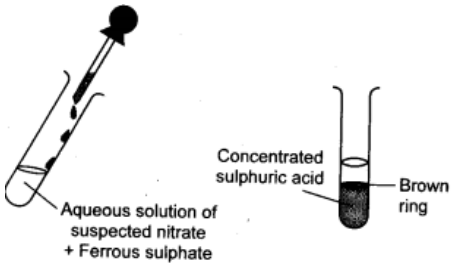
Confirmation of acid radicals

Preparation of sodium carbonate extract (SCE):-

A pinch of salt is mixed with double the amount of sodium carbonate and is boiled with distilled water for sometime. The suspension obtained is filtered.The filtrate is called sodium carbonate extract.

Confirmation of Nitrate (NO_3^-)

1.Copper chips test:- Small amount of the salt in a test tube is heated with conc.sulphuric acid and a few copper chips.	Dark brown fumes of Nitrogen dioxide gas have evolved	Presence of NO_3^- is confirmed.
2. Brown ring test:-		

To the salt solution in a test tube a few ml of freshly prepared ferrous sulphate solution is added then concentrated sulphuric acid is added along the sides of the test tube without shaking .	A brown ring is formed at the junction of layers of acid and the solution. 	Presence of NO_3^- is confirmed.

Identification of basic radicals

Preparation of original solution:-

Small amount of the salt is taken in a test tube and distilled water is added. Shaken well . The clear salt solution is called the original solution.

Analysis of group zero (NH_4^+)

The solid salt is heated with a concentrated solution of sodium hydroxide.	1.No characteristic ammoniacal smell.	Absence of group zero (NH_4^+)

GROUP SEPARATION

Analysis of group I To the original salt solution in a test tube dil.Hydrochloric acid is added.	white precipitate is obtained.	Presence of group I (Pb^{2+})

CONFIRMATION OF BASIC RADICAL - LEAD (Pb^{2+})

1.Potassium Iodide test:- To the original salt solution in a test tube Potassium iodide solution is added.	Yellow precipitate is formed which dissolves in boiling water and on cooling recrystallises.	Presence of lead (Pb^{2+}) is confirmed.
2. Potassium chromate test:- To the original salt solution in a test tube Potassium chromate solution is added.	Yellow precipitate is formed which dissolves in sodium hydroxide solution.	Presence of lead (Pb^{2+}) is confirmed

Result

1. The acid radical present in the given salt is **Nitrate (NO_3^-)**
2. The basic radical present in the given salt is Lead (Pb^{2+}).
3. The given salt is **Lead Nitrate (Pb(NO3)2)**

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